Assignment 3, Problem 1 (part 1)

$y_{i,j} = x_{i,j} + \epsilon_{i,j} \quad \epsilon_{i,j} \sim N(0, \sigma^2)$

$p(x_{i,j} \mid X_{\setminus(i,j)}, Y)$

$x_{i,j} \mid X_{\setminus(i,j)} Y \sim N(\mu_{i,j}, v_{i,j})$,

with

$$v_{i,j} = \frac{1}{\frac{1}{\sigma^2} + \lambda d_{i,j}}, \quad \mu_{i,j} = v_{i,j}\left(\frac{y_{i,j}}{\sigma^2} + \lambda \sum_{(k,l)\in N(i,j)} x_{k,l}\right)$$

$\sigma = 25 \quad \lambda = 0.01$

① Derive expressions for $\mu_{i,j}$ and $v_{i,j}$

By Bayes Rule:

$y_{i,j} = x_{i,j} + \epsilon_{i,j}$

$$P(x_{i,j} \mid X_{\setminus(i,j)} Y) = \underbrace{P(y_{i,j} \mid x_{i,j})}_{\text{data term}} \cdot \underbrace{P(x_{i,j} \mid X_{N(i,j)})}_{\text{prior term}}$$

- Data term:

$$P(y_{i,j} \mid x_{i,j}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_{i,j} - x_{i,j})^2}{2\sigma^2}\right) \propto \exp\left(-\frac{(y_{i,j} - x_{i,j})^2}{2\sigma^2}\right)$$

- Prior term:

$$P(X) \propto \exp\left(-\frac{\lambda}{2} \sum_{(i,j)} \sum_{(k,l)\in N(i,j)} (x_{i,j} - x_{k,l})^2\right)$$

isolate on terms involving $x_{i,j}$:

$$P(x_{i,j} \mid X_{N(i,j)}) \propto \exp\left(-\frac{\lambda}{2} \sum_{(k,l)\in N(i,j)} (x_{i,j} - x_{k,l})^2\right)$$

- So,

$$P(x_{i,j} \mid X_{\setminus(i,j)} Y) \propto \exp\left(\underbrace{-\frac{(y_{i,j} - x_{i,j})^2}{2\sigma^2}} \underbrace{- \frac{\lambda}{2} \sum_{(k,l)\in N(i,j)} (x_{i,j} - x_{k,l})^2}\right)$$

$$\propto -\frac{1}{2\sigma^2}\left(x_{i,j}^2 - 2x_{i,j} y_{i,j} + y_{i,j}^2\right)$$

$$\propto -\frac{x_{i,j}^2}{2\sigma^2} + \frac{y_{i,j}}{\sigma^2} x_{i,j} - \underbrace{\frac{y_{i,j}^2}{2\sigma^2}}_{\text{constant}}$$

$$-\frac{\lambda}{2} \sum_{(k,l)\in N(i,j)} \left(x_{i,j}^2 - 2x_{i,j} x_{k,l} + x_{k,l}^2\right)$$

$$-\frac{\lambda d_{i,j}}{2} x_{i,j}^2 + \lambda x_{i,j} \sum_{k,l} x_{k,l} - \underbrace{\frac{\lambda}{2} \sum_{k,l} x_{k,l}^2}_{\text{constant}}$$

$$\propto -\frac{x_{i,j}^2}{2\sigma^2} - \frac{\lambda d_{i,j}}{2} x_{i,j}^2 + \frac{y_{i,j}}{\sigma^2} x_{i,j} + \lambda x_{i,j} \sum_{k,l} x_{k,l}$$

$$\propto -\frac{1}{2}\left(\frac{1}{\sigma^2} + \lambda d_{i,j}\right) x_{i,j}^2 + \left(\frac{y_{i,j}}{\sigma^2} + \lambda \sum_{k,l} x_{k,l}\right) x_{i,j} \quad \ldots ①$$

Assignment 3, Problem 1 (part 2)

Gaussian $N(\mu, V)$

$$P(x) = \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{(x-\mu)^2}{2V}\right)$$

$$\log P(x) \propto -\frac{(x-\mu)^2}{2V}$$

$$= -\left(\frac{x^2 - 2x\mu + \mu^2}{2V}\right)$$

- Drop constant μ

$$\propto \frac{-x^2}{2V} + \frac{x\mu}{V}$$

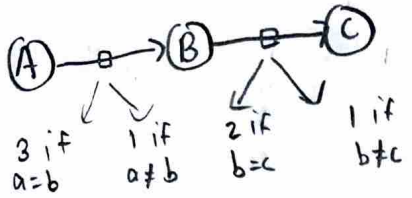
$$= -\frac{1}{2V}x^2 + \frac{\mu}{V}x$$

- Substitute ①

$$\frac{1}{2V_{i,j}} = \frac{1}{2}\left(\frac{1}{\sigma^2} + \lambda d_{i,j}\right) \Rightarrow \boxed{V_{i,j} = \frac{1}{\frac{1}{\sigma^2} + \lambda d_{i,j}}}$$

$$\frac{\mu_{i,j}}{V_{i,j}} = \frac{y_{i,j}}{\sigma^2} + \lambda \sum_{k,l} X_{k,l} \Rightarrow \boxed{\mu_{i,j} = V_{i,j}\left(\frac{y_{i,j}}{\sigma^2} + \lambda \sum_{k,l} X_{k,l}\right)}$$

Assignment 3. Problem 2 (part 1)



$$P(a,b,c) = \frac{1}{Z}\,\psi_{AB}(a,b)\,\psi_{BC}(b,c)$$

1. • $$P(A=1 \mid B=b, C=c) = \frac{P(A=1, B=b, C=c)}{P(B=b, C=c)}$$

$$= \frac{\frac{1}{Z}\psi_{AB}(1,b)\,\psi_{BC}(b,c)}{\sum_a P(A=a, B=b, C=c)} = \frac{\cancel{\frac{1}{Z}}\psi_{AB}(1,b)\,\cancel{\psi_{BC}(b,c)}}{\cancel{\frac{1}{Z}}\psi_{AB}(0,b)\,\cancel{\psi_{BC}(b,c)} + \cancel{\frac{1}{Z}}\psi_{AB}(1,b)\,\cancel{\psi_{BC}(b,c)}}$$

$$= \frac{\psi_{AB}(1,b)}{\psi_{AB}(0,b) + \psi_{AB}(1,b)}$$

a) $$P(A=1 \mid B=0, C=c) = \frac{\psi_{AB}(1,0)}{\psi_{AB}(0,0)+\psi_{AB}(1,0)} = \frac{1}{4} = 0.25$$

b) $$P(A=1 \mid B=1, C=c) = \frac{\psi_{AB}(1,1)}{\psi_{AB}(0,1)+\psi_{AB}(1,1)} = \frac{3}{1+3} = \frac{3}{4} = 0.75$$

• $$P(B=1 \mid A=a, C=c) = \frac{P(B=1, A=a, C=c)}{P(A=a, C=c)} = \frac{\frac{1}{Z}\psi_{AB}(a,1)\,\psi_{BC}(1,c)}{\frac{1}{Z}\psi_{AB}(a,0)\,\psi_{BC}(0,c) + \frac{1}{Z}\psi_{AB}(a,1)\,\psi_{BC}(1,c)}$$

a) $$P(B=1 \mid A=1, C=0) = \frac{\psi_{AB}(1,1)\,\psi_{BC}(1,0)}{\psi_{AB}(1,0)\,\psi_{BC}(0,0) + \psi_{AB}(1,1)\,\psi_{BC}(1,0)} = \frac{3.1}{1.2+3.1} = \frac{3}{5} = 0.6$$

b) $$P(B=1 \mid A=0, C=0) = \frac{\psi_{AB}(0,1)\,\psi_{BC}(1,0)}{\psi_{AB}(0,0)\,\psi_{BC}(0,0) + \psi_{AB}(0,1)\,\psi_{BC}(1,0)} = \frac{1.1}{3.2+1.1} = \frac{1}{7} = 0.143$$

c) $$P(B=1 \mid A=1, C=1) = \frac{3.2}{1.1+3.2} = \frac{6}{7} = 0.857$$

d) $$P(B=1 \mid A=0, C=1) = \frac{1.2}{3.1+1.2} = \frac{2}{5} = 0.4$$

Assignment 3, Problem 2 (part 2)

- $$P(C=1 \mid A=a, B=b) = \frac{P(C=1, A=a, B=b)}{P(A=a, B=b)} = \frac{\cancel{\frac{1}{Z}} \cancel{\psi_{AB}(a,b)}\, \psi_{BC}(b,1)}{\cancel{\frac{1}{Z}} \cancel{\psi_{AB}(a,b)}\, \psi_{BC}(b,0) + \cancel{\frac{1}{Z}} \cancel{\psi_{AB}(a,b)}\, \psi_{BC}(b,1)}$$

$$= \frac{\psi_{BC}(b,1)}{\psi_{BC}(b,0) + \psi_{BC}(b,1)}$$

a) $$P(C=1 \mid A=a, B=0) = \frac{\psi_{BC}(0,1)}{\psi_{BC}(0,0) + \psi_{BC}(0,1)} = \frac{1}{2+1} = \frac{1}{3} = 0.33$$

b) $$P(C=1 \mid A=a, B=1) = \frac{\psi_{BC}(1,1)}{\psi_{BC}(1,0) + \psi_{BC}(1,1)} = \frac{2}{1+2} = \frac{2}{3} = 0.67$$

2. Init state: $(A^0, B^0, C^0) = (0,0,1)$

(a) Sweep 1:

Update A: $P(A=1 \mid B=0, C=1) = 0.25 < u_1 = 0.7 \rightarrow$ set $A=0$

Update B: $P(B=1 \mid A=0, C=1) = 0.4 \geq u_2 = 0.1 \rightarrow$ set $B=1$

Update C: $P(C=1 \mid A=0, B=1) = 0.67 < u_3 = 0.9 \rightarrow$ set $C=0$

$(A^1, B^1, C^1) = (0,1,0)$

(b) Sweep 2:

Update A: $P(A=1 \mid B=1, C=0) = 0.75 \geq u_4 = 0.4 \rightarrow$ set $A=1$

Update B: $P(B=1 \mid A=1, C=0) = 0.6 < u_5 = 0.95 \rightarrow$ set $B=0$

Update C: $P(C=1 \mid A=1, B=0) = 0.33 \geq u_6 = 0.2 \rightarrow$ set $C=1$

$(A^2, B^2, C^2) = (1,0,1)$

Assignment 3, Problem 3

$A \rightarrow B \rightarrow C$

$P(A=1) = 0.4$, $P(A=0) = 0.6$

$P(B=1 \mid A=1) = 0.8$, $P(B=1 \mid A=0) = 0.3$

$P(C=1 \mid B=1) = 0.9$, $P(C=1 \mid B=0) = 0.2$

1) $P(A,B,C) = P(A) \cdot P(B \mid A) \cdot P(C \mid B)$

$P(A=1, B=1, C=1) = P(A=1) \cdot P(B=1 \mid A=1) \cdot P(C=1 \mid B=1)$

$= 0.4 \times 0.8 \times 0.9 = 0.288$

2) $P(C=1) = P(A=0) \cdot P(B=0 \mid A=0) \cdot P(C=1 \mid B=0) +$
$P(A=0) \cdot P(B=1 \mid A=0) \cdot P(C=1 \mid B=1) +$
$P(A=1) \cdot P(B=0 \mid A=1) \cdot P(C=1 \mid B=0) +$
$P(A=1) \cdot P(B=1 \mid A=1) \cdot P(C=1 \mid B=1)$

A	B
0	0
0	1
1	0
1	1

$= [0.6 \times (1-0.3) \times 0.2] + [0.6 \times 0.3 \times 0.9] +$
$[0.4 \times (1-0.8) \times 0.2] + [0.4 \times 0.8 \times 0.9]$

$= 0.084 + 0.162 + 0.016 + 0.288 = 0.55$

3)(a) $P(A) = 0.4 \geq u_1 = 0.3 \rightarrow$ set A to 1

(b) $P(B \mid A=1) \rightarrow P(B=1 \mid A=1) = 0.8 \geq u_2 = 0.75 \rightarrow$ set B to 1

(c) $P(C \mid B=1) \rightarrow P(C=1 \mid B=1) = 0.9 \geq u_3 = 0.5 \rightarrow$ set C to 1

Thus, the sample (A,B,C) is $(1,1,1)$